1. Define a pixel in the context of digital images.
2. What is the difference between grayscale and color image representation?
3. Why is OpenCV's color order BGR instead of RGB?
4. What does the `img.shape` attribute return in NumPy?
5. Define a feature in computer vision.
6. What is a keypoint?
7. State one advantage of ORB over SIFT.
8. What does pooling achieve in CNNs?
9. Define ReLU and explain its purpose.
10. What is dropout and why is it used?
11. State one key contribution of LeNet.
12. What does "compound scaling" mean in EfficientNet?
13. Define viewpoint variation in computer vision.
14. What is occlusion, and why is it challenging?
15. Define a GAN in simple terms.
16. What dataset is used in the notebook?
17. What is the input to the Generator?
18. What does the Discriminator output?
19. Why do we usually keep only the Generator after training?
20. Explain the role of Hue, Saturation, and Value in color representation.
21. What is the purpose of a kernel in image processing?
22. How does histogram equalization enhance image contrast?
23. Explain the concept of descriptors with examples.
24. How does FLANN improve feature matching efficiency?
25. What is the role of RANSAC in robust estimation?
26. Explain the concept of skip connections in ResNet.
27. How does YOLO differ from R-CNN in object detection?
28. What is the role of the encoder and decoder in U-Net?

29. Explain how histogram equalization helps with illumination changes.
30. Explain why normalization of MNIST images is important for GAN training.
31. Describe the two steps in each training round (update D, update G).
32. What is the role of random noise vectors in generating new images?
33. Describe the mathematical operation of convolution and its role in feature extraction.
34. Describe the difference between supervised and unsupervised learning with CV examples.
35. Describe how Vision Transformers use attention to capture global context.
36. Discuss how multi-scale feature extraction addresses scale variation in CNNs.
37. Explain the minimax objective function of GANs.
38. Discuss the three common problems in GAN training: mode collapse, instability, and D too strong.
39. Explain the process of thresholding and its types with examples.
40. Discuss morphological operations (dilation, erosion, opening, closing) with diagrams and applications.
41. Explain Harris Corner Detection — method, pros, cons, and applications.
42. Discuss the role of convolution, pooling, and activation functions in CNNs with examples.
43. Explain the evolution of CNN architectures from LeNet to VGG, highlighting their key innovations.
44. Discuss the importance of ReLU and Dropout in training deep networks.
45. Describe the four core challenges in computer vision (viewpoint, illumination, occlusion, scale) with examples.
46. Describe the analogy of the Generator as a forger and the Discriminator as an inspector.
47. Explain the architecture of the Generator in the notebook (layers, activations, output).
48. Explain the architecture of the Discriminator in the notebook (layers, activations, output).
49. Compare RGB and HSV color models — include their advantages, limitations, and use cases in computer vision.
50. Compare Haar cascades and HOG for face detection — include methodology, strengths, weaknesses, and applications.
51. Compare R-CNN, YOLO, and SSD architectures — include their processes, strengths, weaknesses, and ideal applications.
52. Explain techniques used to overcome these challenges — include data augmentation, histogram equalization, robust feature detection, and multi-scale feature extraction.

53. Discuss the training loop in detail: how labels_real and labels_fake are used, and how gradients flow in each step.

54. Elaborate on contour analysis and object tracking — explain algorithms, challenges (occlusion, scale, clutter), and real-world applications.

55. Elaborate on the Bag of Visual Words pipeline — feature extraction, clustering, histogram representation, and classification.

56. Elaborate on how modern architectures (ResNet, EfficientNet, ViT) solved depth and efficiency challenges, and analyze their impact on real-world vision tasks.

57. Analyze the mathematical foundation of GANs: minimax game, optimal Discriminator, Nash equilibrium, and why the non-saturating generator loss is used.